

SEBASTIAN GUAQUETA JAAR

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EDUCATION

Duke University

Aug 2025 – May 2029

B.S.E. in Biomedical Engineering | B.S.E. in Mechanical Engineering

Durham, NC

Relevant Coursework: Computing for Engineers, Differential Equations, Linear Algebra, Molecular Biology.

Fall 2026 Coursework: Quantitative Physiology, Mechanics of Solids, Fundamentals of Electrical & Computer Engineering.

Palmer Trinity School

Aug 2019 – May 2025

Cum Laude

Miami, FL

ENGINEERING & RESEARCH EXPERIENCE

DukeEngage – Student Engineer

Jun 2026 – Aug 2026

BioSpark Team, Duke–Makerere Partnership

Durham, NC (Remote with Kampala, Uganda)

- Co-developed Neo-Thermo with two Makerere University engineering students in Uganda, creating a ~\$65 early-stage add-on system to monitor and warm gas in neonatal respiratory tubing for resource-limited hospitals.
- Helped review and prioritize ~300 healthcare needs; combined stakeholder interviews, clinical literature, and relevant device standards to define requirements for safety, compatibility, affordability, and local repairability.
- Independently assembled, programmed, and troubleshooted the Duke-side Arduino prototype, integrating two temperature sensors, relay-controlled heating, an OLED display, status LEDs, an audible alarm, and sensor-error detection.
- Conducted preliminary thermal testing and sensor calibration; developed a temperature-estimation model with 0.64°C MAE and 8/10 estimates within $\pm 1^\circ\text{C}$, documenting results and limitations in a technical poster and presentation.

Duke MedDesign – Mechanical Components Team Member

Sep 2025 – Present

Low-Cost Endoscopic Tissue Retraction Device

Durham, NC

- Co-developed CAD models and mechanical drawings for an endoscopic tissue anchor featuring a slider-actuated claw, thumb ring for one-handed operation, and dual off-axis cables for controlled bending.

Duke eNable – Prosthetic Design Team Member

Sep 2025 – Present

Pediatric “Grow-With-Me” Prosthetic Arm Project

Durham, NC

- Developed three candidate concepts for an adjustable forearm-to-wrist interface and benchmarked existing pediatric prosthetic designs and cost-reduction strategies for a device intended to accommodate a child’s growth.

Duke University – EGR 101 Design Team Member

Aug 2025 – Dec 2025

Medication Adherence Alarm for Adults with ADHD

Durham, NC

- Drove concept generation and screening for a medication-adherence wristband, creating CAD models and medium-fidelity prototypes to iterate on form, fit, usability, and component integration.
- Co-developed a BLE/NFC-enabled system linking web scheduling, a pill-bottle module, and wristband alerts; team testing achieved 5/5 alert detection, 20/20 successful NFC scans, 30-meter connectivity, and ~14-day RTC-enabled runtime.

National High School Journal of Science – Published Author

Jun 2024 – May 2025

Nature-Inspired Innovation: Evaluating Biomimicry in Hull Design for Enhanced Engineering Performance

Miami, FL

- Conducted a literature review of 20+ biomimicry applications in engineering, with emphasis on aquatic adaptations.
- Designed and ran experiments testing 4 hull patterns inspired by biological features, analyzing drag reduction.
- Analyzed experimental data and authored a 10-page LaTeX paper published in NHSJS (2025).

ADDITIONAL EXPERIENCE & LEADERSHIP

Duke University – Project Discover Orientation Leader

Apr 2026 – Aug 2026

Experiential Orientation Program

Durham, NC

- Mentored 12 incoming Class of 2030 students during Duke’s experiential orientation program, providing peer guidance on academics and campus life while facilitating group activities and reflective discussions.

GMT Brasseurs RJ / St. Ambroise McAuslan Brewing Co. – Intern

Jun 2023 – Jul 2023

Quality Control, Production, and Logistics

Montreal, Canada

- Supported production and quality control across four product lines, assisting with yeast removal and testing 25+ samples for pH and temperature accuracy; managed inventory and orders for 200+ items and supported retail delivery logistics.

TECHNICAL SKILLS

CAD/Programming: Fusion 360, Onshape, Java, Python, Arduino | **Electronics:** Circuit Assembly, Soldering, Troubleshooting

Testing & Analysis: Sensor Calibration, Thermal Testing, Data Analysis | **Language:** French (Conversational), Creole (Basic)